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0178/01

May/June 2017

1 hour

Candidates answer on the Question Paper.

Additional Materials: Geometrical Instruments
Tracing Paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions.

If working is needed for any question it must be shown below that question.

ELECTRONIC CALCULATORS MUST NOT BE USED FOR THIS PAPER.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 60.

This document consists of **11** printed pages and **1** blank page.



Examinations Council of Lesotho

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[Turn over

1 Work out.

(a) $-12 - 7$

Answer (a) [1]

(b) $3^2 + 5 \times -3$

Answer (b) [2]

(c) $4\frac{2}{5} - 1\frac{7}{8}$

Answer (c) [2]

2

3

6

-8

$\sqrt{10}$

$\frac{22}{7}$

From the list of numbers, write

(a) a composite number,

Answer (a) [1]

(b) a cube number,

Answer (b) [1]

(c) an irrational number.

Answer (c) [1]

- 3 (a) The diagram shows a rhombus.

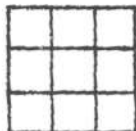


NOT TO
SCALE

Write down the number of lines of symmetry.

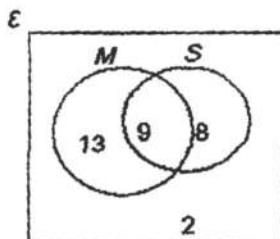
Answer (a) [1]

- (b) The diagram shows a square.



Shade **two** squares in the diagram so that the shape has one line of symmetry and has no rotational symmetry. [1]

- 4 The Venn diagram shows the number of students who study mathematics(M) and science(S) in a class of 32 students.



Find

- (a) $n(M \cap S)$,

Answer (a) [1]

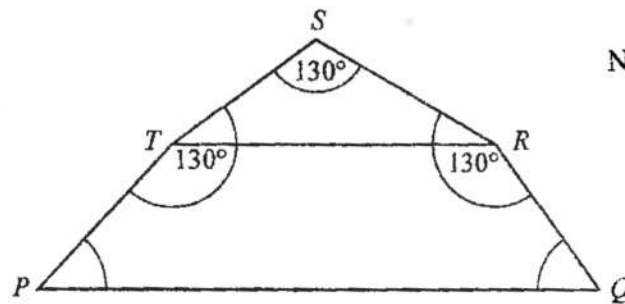
- (b) $n(S)$,

Answer (b) [1]

- (c) the number of students who study maths or science but not both.

Answer (c) [1]

- 5 The diagram shows a pentagon $PQRST$.
 Angles $QRS = RST = STP = 130^\circ$
 Angle $PQR = \text{angle } TPQ$.
 TR is a straight line.



- (a) Find the exterior angle T .

Answer (a) $^\circ$ [1]

- (b) Calculate the angle

- (i) SRT ,

Answer (b)(i) [2]

- (ii) PQR .

Answer (b)(ii) $^\circ$ [3]

6 Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, find

(a) $2\mathbf{a} + \mathbf{b}$,

Answer (a) [2]

(b) $\mathbf{a} - \mathbf{b}$,

Answer (b) [1]

(c) the magnitude of $\mathbf{b}$.

Answer (c) [2]



7 (a) Evaluate

(i) $2^0 + 3^2$

Answer (a)(i) [1]

(ii) $\left(\frac{1}{5}\right)^{-2}$

Answer (a)(ii) [2]

(b) Solve

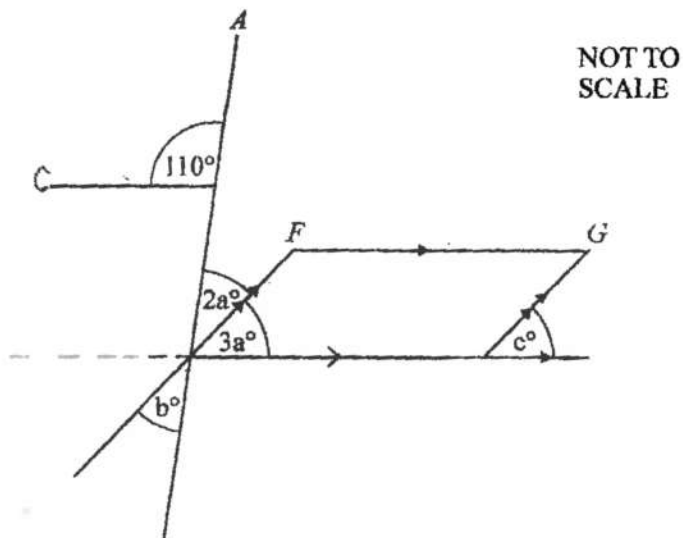
(i) $2^a \times 2^3 = 32,$

Answer (b)(i) $a =$ [1]

(ii) $\frac{4x}{3} - 8 = 6 - x.$

Answer (b)(ii) $x =$ [1]

- 8 *OFGH* is a parallelogram.
EOF is parallel to *HG* and *CD* is parallel to *FG*.
 Angle $CDA = 110^\circ$.



Calculate the value of

(a) a ,

Answer (a) $a = \dots\dots\dots$ [2]

(b) b ,

Answer (b) $b = \dots\dots\dots$ [1]

(c) c .

Answer (c) $c = \dots\dots\dots$ [1]

9 (a) Write 0.0539 correct to

(i) the nearest thousandth,

Answer (a)(i) [1]

(ii) two significant figures.

Answer (a)(ii) [1]

(b) The diameter of Jupiter is 139 822 km.

(i) Write the number in words.

Answer (b)(i)

..... [1]

(ii) Find the radius of Jupiter in metres.
Give your answer in standard form.

Answer (b)(ii) m [2]

(c) By making suitable approximations, estimate the value of

$$\frac{503.6 \times (27.33)^2}{28.03}$$

Answer (c) [2]

10 $r = \frac{q(p - q)}{3}$

- (a) Find r when $p = 2$ and $q = -6$.

Answer (a) $r = \dots\dots\dots$ [2]

- (b) Make p the subject of the formula.

Answer (b) $p = \dots\dots\dots$ [2]

- 11 The table shows the opening and closing hours of a local supermarket.

	Mon – Fri	Saturday	Sunday
Opening	07 15	07 45	08 15
Closing	19 45	19 45	18 30

- (a) How many more hours is the supermarket open on Monday than Sunday?

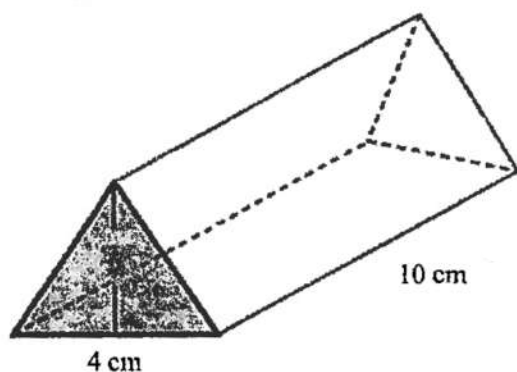
Answer (a) $\dots\dots\dots$ [2]

- (b) Calculate the total number of hours the supermarket is open per week.

Answer (b) $\dots\dots\dots$ [2]



- 12 The diagram shows a triangular prism with length 10 cm.
The shaded cross section is a triangle of base 4 cm and perpendicular height h .



NOT TO
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The volume of the prism is 100 cm^3 .

- (a) Calculate the area of the shaded cross section.

Answer (a) cm^2 [2]

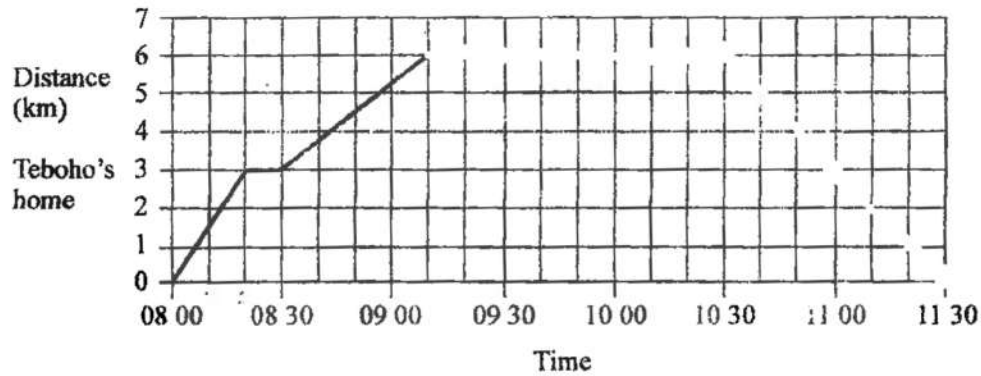
- (b) Calculate the height h .

Answer (b)(i) cm [2]

Give your answer to **part (b)(i)** in millimetres.

Answer (b)(ii) mm [1]

- 13 The travel graph shows Teboho's journey, by bicycle, from his home to the clinic.



- (a) Teboho stopped at a shop on the way to the clinic.
For how many minutes did he stop?

Answer (a) min [1]

- (b) Write down the time he arrived at the clinic.

Answer (b) [1]

- (c) Teboho stayed at the clinic for 80 minutes and then cycled home without stopping.
His return journey took 1 hour.

Complete the travel graph.

[2]